$q = 1.6 \cdot 10^{-19}\,\mathrm{A}s \qquad k = 1.38 \cdot 10^{-23}\,\frac{\mathrm{J}}{\mathrm{K}} = 8.625 \cdot 10^{-5}\,\frac{\mathrm{eV}}{\mathrm{K}} \qquad \varepsilon_0 = 8.854 \cdot 10^{-14}\,\frac{F}{cm} \qquad \varepsilon_{rSi} = 11.7$

$U_T = \frac{kT}{q} \qquad n_i = 3.07 \cdot 10^{16} T^{\frac{3}{2}} e^{-\frac{6937}{T}} \qquad n \cdot p = {n_i}^2 \qquad n + N_A = p + N_D \qquad U_K = U_T \ln \frac{N_A N_D}{n_i^2}$

$J_s = q \cdot n_i^2 \cdot \left(\frac{L_n}{N_A \tau_n} + \frac{L_p}{N_D \tau_p} \right) = q \cdot n_i^2 \cdot \left(\frac{D_n}{N_A L_n} + \frac{D_p}{N_D L_p} \right) = q \cdot n_i^2 \cdot \left(\frac{1}{N_A} \sqrt{\frac{D_n}{\tau_n}} + \frac{1}{N_D} \sqrt{\frac{D_p}{\tau_p}} \right)$

$J = q \left(\frac{D_n n_{0p}}{L_n} + \frac{D_p p_{0n}}{L_p} \right) \left(e^{\frac{U}{U_T}} - 1 \right) = J_s \left(e^{\frac{U}{U_T}} - 1 \right) \qquad I = I_s \left(e^{\frac{U}{mU_T}} - 1 \right)$

npn tranzistor:

$I_E = I_{nE} + I_{pE} + I_{rg} \qquad I_B = I_{pE} + I_{rg} + I_{nE} - I_{nC} - I_{CB0} \qquad I_C = I_{nC} + I_{CB0}$

$\gamma = \frac{I_{nE}}{I_E} \qquad \beta^* = \frac{I_{nC}}{I_{nE}} \qquad \alpha = \gamma \beta^* \qquad \beta = \frac{\alpha}{1 - \alpha}$

$h_{fe} = \frac{\beta}{1 + j\omega r_{b'e}(C_e + C_c)} \qquad f_T = \beta \cdot f_\beta$

JFET:

$I_D = G_0 \left[1 - \sqrt{\frac{2\varepsilon}{qNa^2}(U_K - U_{GS})} \right] U_{DS}$

$I_D = G_0 \left[U_{DS} - \frac{2}{3} \frac{1}{\sqrt{U_K - U_{GS0}}} \left[(U_K - U_{GS0} + U_{GS})^{\frac{3}{2}} - (U_K - U_{GS})^{\frac{3}{2}} \right] \right]$

$I_{DS} = G_0 \left[U_{GS} - U_{GS0} - \frac{2}{3}(U_K - U_{GS0}) \left[1 - \left(\frac{U_K - U_{GS}}{U_K - U_{GS0}} \right)^{\frac{3}{2}} \right] \right] \qquad I_{DS} = I_{DSS} \left(1 - \frac{U_{GS}}{U_{GS0}} \right)^2$

$g_m = \frac{G_0}{\sqrt{U_K - U_{GS0}}} \left[\sqrt{U_K - U_{GS} - U_{GS0}} - \sqrt{U_K - U_{GS}} \right] \qquad g_m = 2\sqrt{K \cdot I_D} \qquad K = \frac{I_{DSS}}{U_{GS0}^2}$

$r_d = \frac{1}{G_0 \left[1 - \sqrt{\frac{U_K - U_{GS} + U_{DS}}{U_K - U_{GS0}}} \right]} \qquad g_m = \frac{\Delta I_D}{\Delta U_{GS}} \qquad r_d = \frac{\Delta U_{DS}}{\Delta I_D}$

MOSFET:

$I_D = K(U_{GS} - U_{GS0}) U_{DS} \qquad K = \mu \frac{\varepsilon_{ox}}{d_{ox}} \frac{W}{L} \qquad \varepsilon_{rox} = 3.82$

$I_D = K \left[(U_{GS} - U_{GS0}) U_{DS} - \frac{U_{DS}^2}{2} \right]$

$I_{DS} = \frac{K}{2}(U_{GS} - U_{GS0})^2 \qquad I_{DS} = I_{DSS} \left(1 - \frac{U_{GS}}{U_{GS0}} \right)^2$

$g_m = K \cdot U_{DS} \qquad g_m = \sqrt{2K \cdot I_D} \qquad K = \frac{2 I_{DSS}}{U_{GS0}^2}$

$r_d = \frac{1}{K(U_{GS} - U_{GS0} - U_{DS})} = \frac{1}{\lambda I_{DSS} \left(1 - \frac{U_{GS}}{U_{GS0}} \right)^2}$